



Project 2

GROUP C

Dynamic Finite Element Analysis of Structure ME-473

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Material	E (GPa)	Density (kg/m ³)	Cross-Sectional Area
Structural steel	200	7850	15–50 cm ²
Aluminum alloy	69	2700	20–70 cm ²
Engineered wood	12	600	30–100 cm ²

Table 1: Material Properties

1 Introduction

Finite Element Method analysis of trusses is a useful tool for evaluating bridge designs. In this project report, three different 2D truss bridge designs are created using three different materials. The design space and load are imposed and are the same for all three designs. Specific requirements for the static and dynamic behavior are to be fulfilled. The created designs are analyzed statically and dynamically using the Finite Element Method, the requirements are verified, and the results are discussed.

1.1 Problem

The bridges have two pairs of anchor points that are 20 meter apart. A vertical load is applied at the center of this span, although the height of the point of attack can be chosen freely. The anchor points on the left and right are spaced 1 meter apart (see Figure 1).

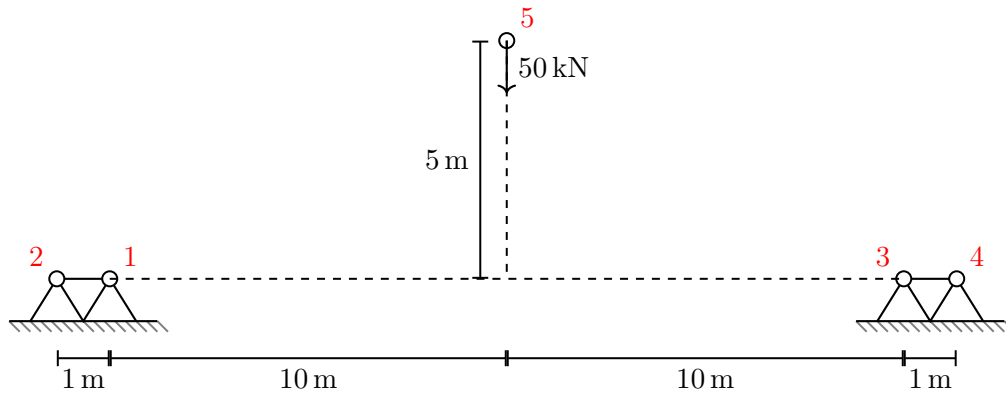


Figure 1: Problem Specification

Three materials, structural steel, aluminum alloy, and engineering wood, are to be used. A unique design must be created for each of these materials. Their material properties and the allowed cross-sections of the bars are given in Table 1.

Static and dynamic analysis is done on all of the designs. The designs have to meet the following requirements:

1. No bar shall yield in the static analysis
2. The displacement at each node shall be kept below 40 mm in the static analysis

3. The total material volume shall not exceed 1m^3 for structural steel, 1.7m^3 for aluminum alloy and 2m^3 for engineering wood
4. The first natural frequency shall not be between 1.5 and 10 Hz.

If possible, a first natural frequency above 50 Hz should be achieved.

2 Methodology

2.1 2D Trusses

2D trusses rely on assumptions that greatly simplify the complexity of calculations. This makes them a useful tool for studying specific types of structures.

2.1.1 Assumptions

2D trusses consist of bars that all lie in a common plane and whose ends are connected through frictionless joints, meaning that they cannot transfer torques. External loads can only be applied at the joints, and their movement is limited to the 2 dimensions of the plane. The composing bars are subject only to compressive or tensile forces and thus can only deform in the axial direction. Their weight can be neglected. In addition, the bars are made of isotropic linear elastic materials, and their cross-sectional profile and material properties do not vary over the length of the bar.

2.1.2 Bar Model

The above assumptions allow for the mechanical model pictured in Figure 2. The model of a bar is fully characterized by its material parameters, which are the Young's modulus E , the cross-sectional area A and the density ρ , as well as the length ℓ and the applied loads at the ends of the bar in the axial direction, f_1 and f_2 .

Strong Form

The strong form of the model then gives us equations for the axial displacement $u(x', t)$:

$$EA\partial_{x'}^2 u(x', t) = \rho A\ddot{u}(x', t) \quad \forall (x', t) \in]0, \ell[\times]0, T[\quad (1)$$

$$EA\partial_{x'} u(0, t) = -f_1(t) \quad \forall t \in]0, T[\quad (2)$$

$$EA\partial_{x'} u(\ell, t) = f_2(t) \quad \forall t \in]0, T[\quad (3)$$

$$u(x', 0) = u_0(x') \quad \forall x' \in]0, \ell[\quad (4)$$

$$\dot{u}(x', 0) = v_0(x') \quad \forall x' \in]0, \ell[\quad (5)$$

(1) can be deduced from observing an infinitesimal slice dx' of the bar. It follows from applying the second law of motion on the slice that $A\partial_{x'}\sigma = A\rho\ddot{u}$. Using the strain relation $\epsilon = \partial_{x'}u(x', t)$ and Hooke's law $\sigma = E\epsilon$, we find (1). (2) and (3) follow from the forces acting on both ends of the bar and again the strain relation and Hooke's law. (4) and (5) are the initial conditions of the problem.

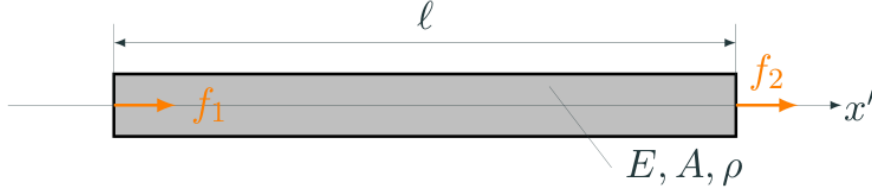


Figure 2: Model parameters of a bar [1]

Weak Form

To derive the weak form we introduce the virtual displacement δu and impose that

$$u \in U = \{u(x', t) \in H^1(]0, \ell[)\} \quad (6)$$

$$\delta u \in V = \{\delta u \in H^1(]0, \ell[)\} \quad (7)$$

with the Sobolev space $H^1(\Omega)$. Multiplying (1) by δu and integrating over $]0, \ell[$ one obtains:

$$\int_0^\ell \delta u E A \partial_{x'} u(x', t) dx = \int_0^\ell \delta u \rho A \ddot{u}(x', t) dx \quad \forall \delta u \quad (8)$$

The first term is integrated by parts:

$$[\delta u E A \partial_{x'} u]_0^\ell - \int_0^\ell E A (\partial_{x'} u) (\partial_{x'} \delta u) dx = \int_0^\ell \rho A \ddot{u} \delta u dx \quad (9)$$

Using (2) and (3):

$$\int_0^\ell E A (\partial_{x'} u) (\partial_{x'} \delta u) dx + \int_0^\ell \rho A \ddot{u} \delta u dx - \delta u(0) f_1 - \delta u(\ell) f_2 = 0 \quad (10)$$

Using the same approach on the initial conditions yields:

$$\int_0^\ell \rho \delta u u|_{t=0} dx = \int_0^\ell \rho \delta u u_0 dx, \quad (11)$$

$$\int_0^\ell \rho \delta u \dot{u}|_{t=0} dx = \int_0^\ell \rho \delta u v_0 dx, \quad (12)$$

A note on boundary conditions for bars within the truss: (10) applies in the general case where no displacement boundary condition is imposed on the bar. However, if the bar is connected to a fixed pin at one of its ends (e.g., at $x = 0$), then that end cannot undergo any displacement, leading to $\delta u(0) = 0$. It is important to note that such conditions are not accounted for at this stage, as all bars in the truss are currently assumed to be free to move at both ends. Displacement boundary conditions are introduced later during the global truss assembly.

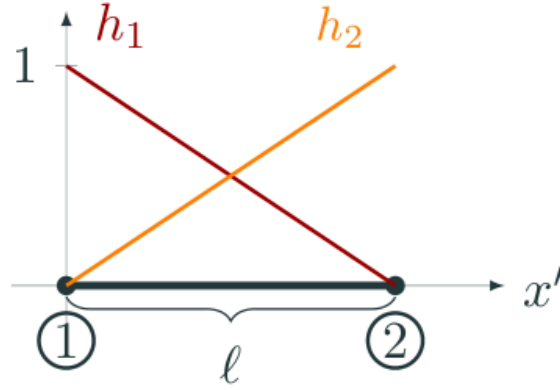


Figure 3: Bar linear local shape functions [1]

As for the force boundary conditions (f_1 and f_2), they are defined by the force contributions from other bars connected to the same pins, based on equilibrium in the context of static analysis. In other words, for each unconstrained pin, the sum of internal forces from connected bars must be equal and opposite to the external force applied at the pin (if any), or zero if no external force is present.

Galerkin Method

Using the Galerkin method, we approximate the (virtual) displacement using shape functions $\mathbf{H}(x) = [h_1(x') \quad h_2(x')]$ and time dependent function $\vec{q}(t) = [q_1(t), q_2(t)]^T$:

$$u^h(x', t) = \mathbf{H}(x) \vec{q}(t) \quad (13)$$

$$\delta u^h(x') = \mathbf{H}(x) \delta \vec{q}(t) \quad (14)$$

For the 2D non-oriented bar, we use simple linear local shape functions (see Figure 3):

$$h_1(x') = 1 - \frac{x'}{\ell} \quad (15)$$

$$h_2(x') = \frac{x'}{\ell} \quad (16)$$

Inserting the approximated displacements into the weak form equation yields the semi-discrete weak form for a bar element:

$$\delta \vec{q} \cdot \left(\int_0^\ell EA (\nabla \mathbf{H})^T \nabla \mathbf{H} dx \right) \vec{q} + \delta \vec{q} \cdot \left(\int_0^\ell \rho A \mathbf{H}^T \mathbf{H} dx \right) \ddot{\vec{q}} - \delta \vec{q} \cdot (\mathbf{H}^T(0) f_1 + \mathbf{H}^T(\ell) f_2) dx = 0 \quad (17)$$

which can be rewritten as follows, dividing out $\delta \vec{q}$:

$$\mathbf{K} \vec{q}(t) + \mathbf{M} \ddot{\vec{q}}(t) - \vec{f}(t) = 0 \quad (18)$$

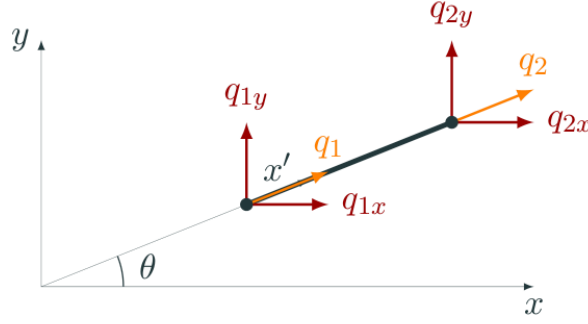


Figure 4: Local bar coordinates q_1 and q_2 and global bar coordinates q_{1x} , q_{1y} and q_{2x} , q_{2y} .

where the stiffness matrix, $\mathbf{K}$, the mass matrix $\mathbf{M}$ and the load vector $\vec{f}$ are given by:

$$\mathbf{K} = \int_0^\ell EA \begin{bmatrix} (h'_1)^2 & h'_1 h'_2 \\ h'_2 h'_1 & (h'_2)^2 \end{bmatrix} dx' = \frac{EA}{\ell} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad (19)$$

$$\mathbf{M} = \int_0^\ell \rho A \begin{bmatrix} (h_1)^2 & h_1 h_2 \\ h_2 h_1 & (h_2)^2 \end{bmatrix} dx' = \frac{\rho A \ell}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad (20)$$

$$\vec{f}(t) = \begin{bmatrix} h_1(0) \\ h_2(0) \end{bmatrix} f_1(t) + \begin{bmatrix} h_1(\ell) \\ h_2(\ell) \end{bmatrix} f_2(t) = \begin{bmatrix} f_1(t) \\ f_2(t) \end{bmatrix} \quad (21)$$

where h'_1 and h'_2 are the derivatives of h_1 and h_2 with respect to x' .

2.1.3 From bar to truss

We assemble our Finite Element model from the bar elements described above. For this, several steps are still required.

Transformation from local to global coordinates

The first step in assembling a Finite Element truss model is transforming the local bar element parameters to the global coordinate system of the truss. From now on, the subscript $\cdot_{loc}$ refers to the local non-oriented parameters presented above.

A transformation matrix $\vec{q}_{loc} = \mathbf{T}\vec{q}$ is derived from Figure 4, yielding [1]:

$$\begin{bmatrix} q_1 \\ q_2 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 \\ 0 & 0 & \cos(\theta) & \sin(\theta) \end{bmatrix} \begin{bmatrix} q_{1x} \\ q_{1y} \\ q_{2x} \\ q_{2y} \end{bmatrix} \quad (22)$$

The stiffness and mass matrices as well as the load vector are transformed:

$$\mathbf{K} = \mathbf{T}^T \mathbf{K}_{loc} \mathbf{T} \quad (23)$$

$$\mathbf{M} = \mathbf{T}^T \mathbf{M}_{loc} \mathbf{T} \quad (24)$$

$$\vec{f} = \mathbf{T}^T \vec{f}_{loc} \quad (25)$$

The global matrices are of size 4×4 and the load vector of size 4×1 . Their full form is not printed here for brevity.

Assembly

Next, the $2n \times 2n$ global stiffness and mass matrices are created respectively, as well as a global loads vector of size $2n \times 1$, where n denotes the number of nodes. A connectivity table is set up, relating which nodes (joints) belong to which element. From now on, $^e \cdot$ signifies the element. The values of the transformed element stiffness matrices $^e \mathbf{K}$, mass matrices $^e \mathbf{M}$ and load vectors $^e \vec{f}$ are inserted into the global matrices at the right positions according to the connectivity table, yielding:

$$\mathbf{K} = \mathbf{A}_e^e \mathbf{K} \quad (26)$$

$$\mathbf{M} = \mathbf{A}_e^e \mathbf{M} \quad (27)$$

$$\vec{f} = \mathbf{A}_e^e \vec{f} \quad (28)$$

The rows and columns relating to constrained nodes, meaning nodes whose displacement is imposed to be zero, are removed, leaving reduced matrices $\mathbf{K}_f$, $\mathbf{M}_f$ and $\vec{f}_f$ of free nodes. This yields the final equation for the displacement of the free nodes of the truss:

$$\mathbf{M}_f \ddot{\vec{q}}_f + \mathbf{K}_f \vec{q}_f = \vec{f}_f \quad (29)$$

2.1.4 Static Analysis

The static analysis of the truss is necessary to check that the displacement of the nodes under load does not exceed a specified limit. Also, the stresses in the bars may not exceed the range of elastic material behavior. This makes sure that the structure can take these loads for an indefinite amount of time.

The static model ignores transient effects, which also means that the applied forces need to be constant. The truss model simplifies to:

$$\mathbf{K}_f \vec{q}_f = \vec{f}_{f,max} \quad (30)$$

This is a simple linear system of equations that can be solved quickly by optimized algorithms, especially when $\mathbf{K}_f$ is a band matrix. q_f is subsequently checked against the allowed displacement.

To check the bar stresses, we re-transform the global displacements to the local elements using the connectivity table and apply Hooke's law. Care has to be taken to reintroduce the constraint degrees of freedom before the transformation. Then, the local element stresses are given by:

$${}^e\sigma^h = \frac{{}^eE}{{}^e\ell} \begin{bmatrix} -\cos({}^e\theta) & -\sin({}^e\theta) & \cos({}^e\theta) & \sin({}^e\theta) \end{bmatrix} \begin{bmatrix} q_{ix} \\ q_{iy} \\ q_{jx} \\ q_{jy} \end{bmatrix} \quad (31)$$

2.1.5 Dynamic Analysis

To analyze the transient behavior of the system under dynamic loads, we first solve the eigenvalue problem resulting from the ansatz $\vec{q}_f = \Phi e^{i\omega t}$:

$$(\mathbf{K}_f - \omega^2 \mathbf{M}_f) \Phi = \mathbf{0} \quad (32)$$

Solving for the eigenvalues ω^2 and the resulting natural frequencies $f = \frac{\omega}{2\pi}$, we gain a first insight into the dynamic behavior of the system. The eigenvectors contained in Φ are the corresponding mode shapes.

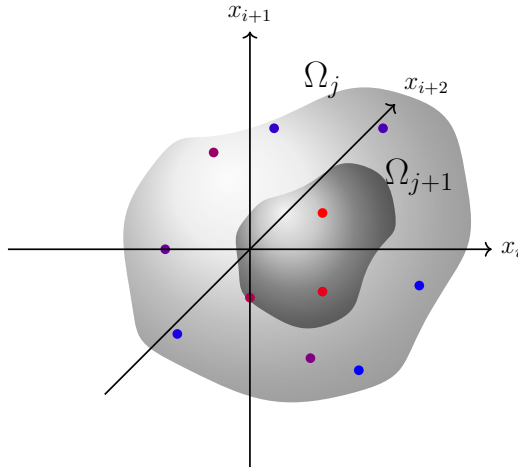
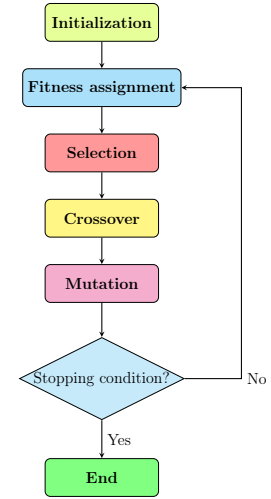
2.2 Optimization

Once the finite element code was developed, it was only a matter of finding the best geometry that could meet the construction requirements imposed by the project.

Even though this was a fairly simple task that could be executed by hand, it was considered to be more formative to write an optimizer to quickly find the best configuration for different types of bridges and materials, that could match the requirements. For this analysis, a load height of $a = 5$ m was assumed. Refer to figure 1 for the geometric representation.

Due to the non-convexity of the problem, many traditional optimizers might have struggled to find the best design (e.g. gradient descent method), hence a Genetic Algorithm approach was followed; see figure 5b. This type of optimization algorithm is inspired by the evolutionary theory in nature. It's based on the generation of random populations that can be evaluated by a scalar fitness function. The best individuals from each random generation are selected to "reproduce" based on their fitness. Practically, meaning that the boundaries of the input variable are restricted around the best individuals, and the next generation is generated from this subset of the original domain, fig. 5a.

This iterative loop is then interrupted when a maximum number of generations (user-defined variable) is reached or when the evolution of the element reaches a stall and doesn't change anymore.

(a) Example of domain restriction at generic step j 

(b) Generic genetic algorithm

Note: i represent a generic coordinate in the $\mathbf{R}^N$ space determined by the decision variable of the problem.

This approach views the problem as a black box and enables the identification of a local minimum of the Cost Function within the boundary imposed for each variable, thereby allowing for the discovery of a solution, a local minimum, in all cases independently of the problem being solved.

Nevertheless, this approach is highly computationally expensive as each generation might have several hundred individuals, and the optimizer might run for tens of iterations. Hence, the problem might need to be simplified and the code optimized not require too much time to run.

The genetic algorithm used in the code is the `ga`¹ implemented in MATLAB[®] in the optimization toolbox².

2.2.1 Problem definition

The optimizer was developed to receive as input a certain "reference" bridge geometry, which is modified by changing the position of each node around the original position (here defined by the subscript *reference*) into a domain defined as:

$$\begin{aligned} x_{reference}^n - \epsilon &\leq x_i^n \leq x_{reference}^n + \epsilon \\ y_{reference}^n - \epsilon &\leq y_i^n \leq y_{reference}^n + \epsilon \end{aligned} \quad (33)$$

Where ϵ is set by the user and it is an arbitrary real positive number ($\epsilon \in \mathbb{R}^+$). Note the superscript n , represents the nodes.

The smaller ϵ is, the faster the optimizer is, as the local minimum is searched into a smaller domain. Nevertheless, it might find solutions that do not differ from the original by much, hence the quality might be limited. Higher values theoretically lead to a better minimum to

¹<https://www.mathworks.com/help/gads/ga.html>

²<https://www.mathworks.com/products/optimization.html>

be found, but it is important that neighbor node do not cross over, as the definition of the stiffness and mass matrices would become impossible, as sine and cosine function used into the matrices could not be defined as the length of some elements would be null.

Moreover from an engineering point of view, it would not make sense to treat the position of the nodes as continuous function, as in reality, the geometrical definition of such big structures is never defined at the centimeter scale, neither at the millimeter. Thus, the domain into which each node can move (eq. 36) is discretized with a certain square meshgrid of finite size, which represents the only sites in which the node can move. In particular, in the code, the mesh size was assumed to be 10 cm, so the nodal position resolution is defined as follows:

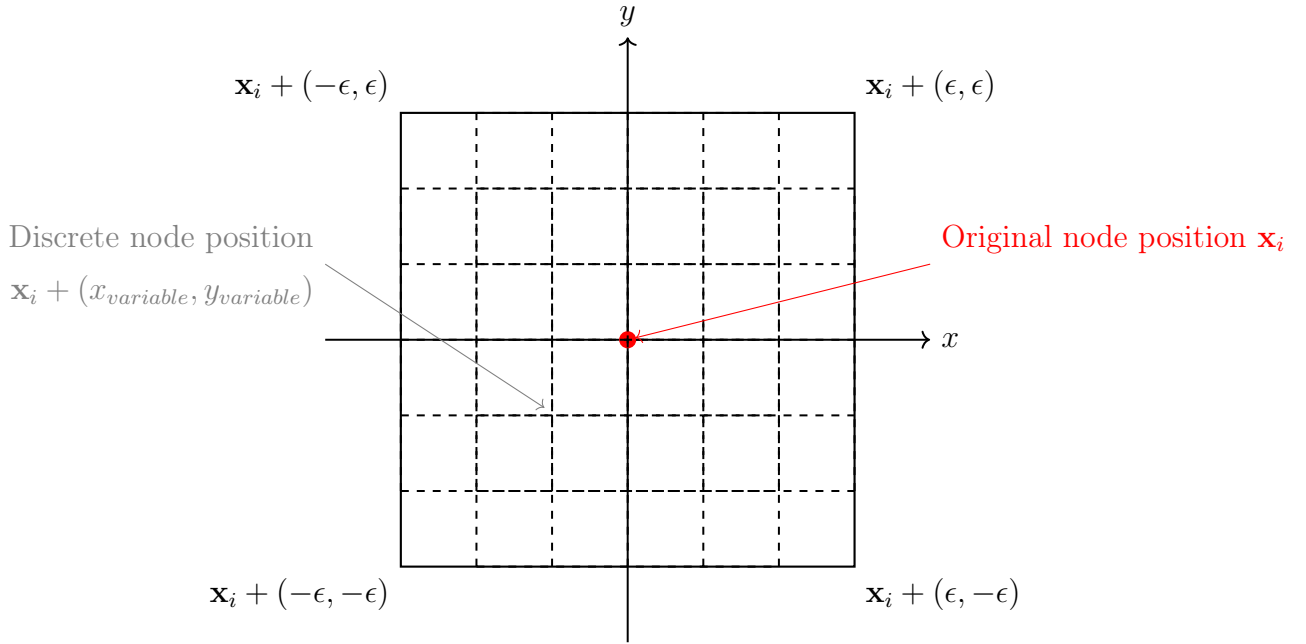


Figure 6: Discretization for nodal position around original position, for a generic node

Furthermore, the optimizer is set to optimize the bridge, changing the area of each truss within the boundary set by the problem:

$$A_{LB}^{material} \leq A_i^e \leq A_{UB}^{material} \quad (34)$$

Where A_i^e represents the area of a generic element i , while the subscripts LB and UP , represent Lower and Upper Boundary for a certain material, given by the problem.

At last, the problem definition allows for some geometrical constraints that can help simplify the problem:

- Some nodes, given by the problem, are fixed and cannot be moved, thus their coordinates are not considered in the decision variables vector;
- There's a central symmetry:

- Nodes on the symmetry axis can only be moved in the vertical position, obtained by setting Lower and Upper boundaries for the horizontal position to be the same as the original node position;
- It is possible to define most variables only once, as the nodes coordinate and truss area must be the same on both sides of the bridge;
- Nodes on the road surface can only move horizontally, to grant continuity in pavement, obtained by setting Lower and Upper boundaries for the vertical position to be the same as the original node position.

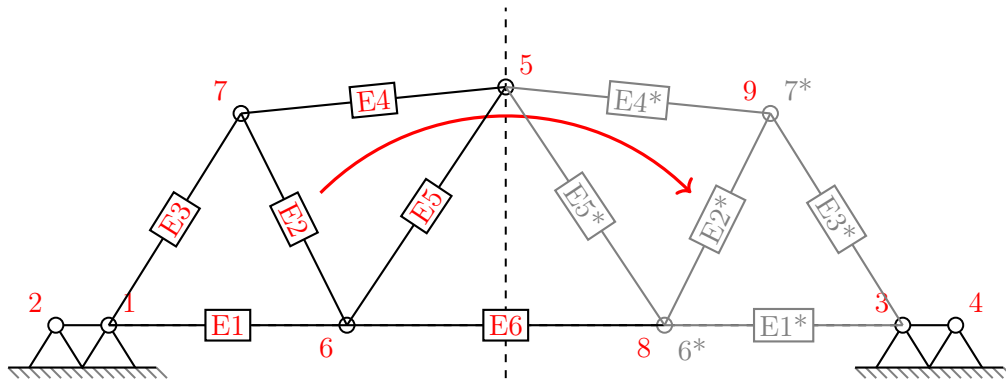


Figure 7: Example of symmetry constraint representation

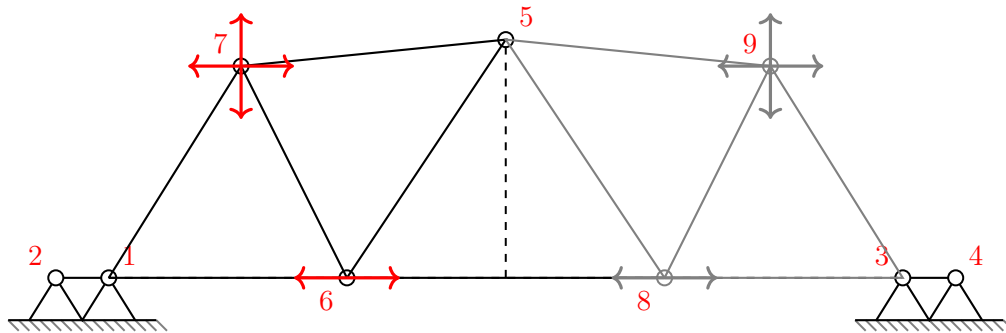


Figure 8: Example of nodal mobility, with problem constraints

Thus, the optimizer is set to optimize a column vector, which represents in the first positions all the x and y coordinates for the free nodes, followed by all the areas of the trusses. Lower and Upper boundaries are represented by two column vectors, similarly defined.

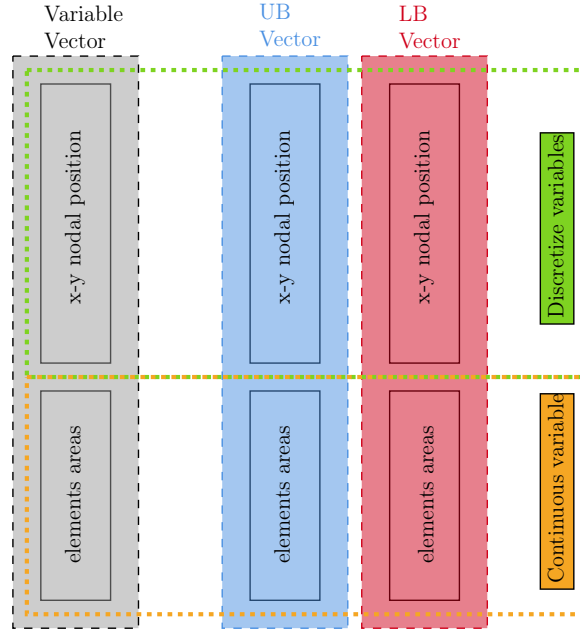


Figure 9: Optimizer variable vectors

2.2.2 Fitness Function

At last, it is important to define a Fitness Function, which is used to evaluate the individuals in each iterative generation.

Each individual, defined by the input decisional variable vector, represents a different bridge. For each individual:

- Stiffness and Mass matrices are computed;
- Volume of all trusses is evaluated (V_{Total}), as one of the problem constraints was to use a limited amount of material;
- Modal analysis is carried out to determine the first modal frequency (f_{min});
- Static analysis is carried out to determine:
 - displacement at each node (q_i^n);
 - max displacement in the structure (q_{max});
- From the result of the static analysis, stresses (ε_i^e) are computed for the loaded structure, and max stress is evaluated (ε_{max}).

Eventually, the fitness function is computed as follows:

$$\mathcal{F}_{cost} = \frac{V_{Total}}{V_{lim}} W_{volume} + \frac{\text{mean}(q_i^n)}{q_{lim}} W_q - \frac{f_{min}}{f_{lim}} W_{freq} - \frac{\text{mean}(\varepsilon_i^e)}{\varepsilon_{lim}} W_{stress} \quad (35)$$

In the cost function, we can find 4 contributions, since the optimizer is set to minimize the overall value of this function, the first 2 are to be minimized as well, which represent the Volume and the average displacement in all nodes. On the contrary, the last two are to be maximized,

which represent the first modal frequency and the average stresses in the all structure, in order to have well-used material.

Each value is normalized in respect to the limits set by the problem (V_{lim} for limit volume, q_{lim} for limit displacement, f_{lim} for limit in the frequency, and ε_{lim} for yield stress of the material). All normalized values are multiplied by some weights, W , which might be needed to stress the importance of one parameter over the others, but in this case were kept equal to the unitary value.

Furthermore, some penalties are applied to the fitness function if one or more of the limits set by the problem are exceeded. These penalties are in the form of exponential functions to quickly diverge and drive the optimization to values within the boundaries.

$$\mathcal{F}_{cost} = \mathcal{F}_{cost} + \sum_i^4 penalty_i$$

$$\begin{aligned} penalty_1 &= \exp\left(\frac{V_{Total}}{V_{lim}}\right)W_{freq} & for \ V_{Total} > V_{lim} \\ penalty_2 &= \exp\left(\frac{q_{max}}{q_{lim}}\right)W_q & for \ q_{max} > q_{lim} \\ penalty_3 &= \exp\left(\frac{f_{lim}}{f_{min}}\right)W_{freq} & for \ f_{min} < f_{lim} \\ penalty_4 &= \exp\left(\frac{\varepsilon_{max}}{\varepsilon_{lim}}\right)W_{stress} & for \ \varepsilon_{max} > \varepsilon_{lim} \end{aligned} \quad (36)$$

2.2.3 Reference geometry

As mentioned, the optimizer was developed to improve a reference bridge geometry to best match the boundary set from the problems. In particular, different starting geometries were defined to be optimized. It's important to note that for all reference geometries the initial elements' areas were defined as a random value that lies inside the areas boundary of individual materials (user-defined variable).

In short a Warren bridge design was used as reference geometry for the structural steel bridge, while a Pratt bridge design was used for the aluminum bridge. Lastly a warren bridge inspired design was also used for the reference geometry for the engineering wood bridge, but due to the relatively low stiffness of the material, additional reinforcement were added as well. Further details on the reference bridge designs are presented in appendix A.1.

3 Results

3.1 Structural steel

The final iteration of the optimizer gave us the geometric parameters presented on figure 10. Furthermore tables 15 and 16 in appendix A.2 present the nodal coordinates, connectivity table and cross-sectional areas for the structure. We found that the yield stress for structural steel was $\sigma_y = 250$ MPa [2]. The total amount of steel used to build this truss is 0.196 m^3 , which is less than 1 m^3 and therefore fully meets the volume requirement.

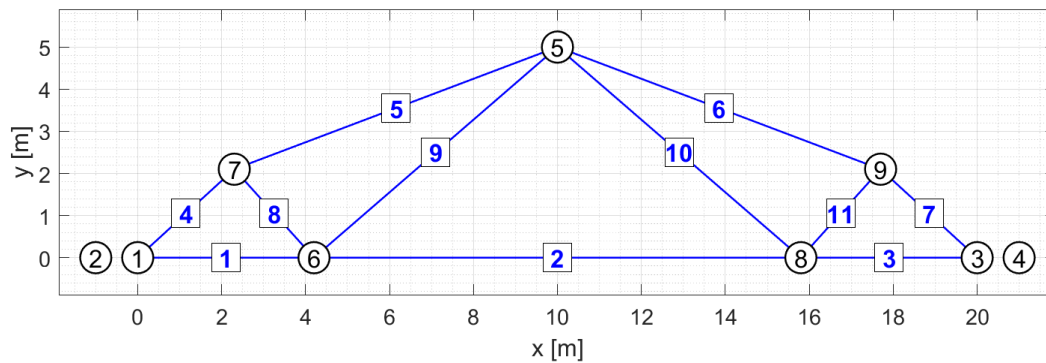


Figure 10: Optimized structure for the structural steel bridge.

3.1.1 Static analysis

Table 2 shows the displacements found for individual nodes, while table 3 shows the stresses found for individual elements. Due to symmetry, results are only shown for the left side of the bridge. Results are not shown for nodes 1 and 2, as they are constrained in both degrees of freedom. A graphical representation of the deformation is shown on figure 11.

Node number	5	6	7
Horizontal displacement [mm]	0.000	-0.079	0.018
Vertical displacement [mm]	-1.212	-0.465	-0.202
Total displacement [mm]	1.212	0.471	0.202

Table 2: Static displacements of the steel bridge.

Element number	1	2	4	5	8	9
Stress [MPa]	-3.74	2.71	-7.89	-9.05	9.23	-11.18

Table 3: Static stresses in steel bridge.

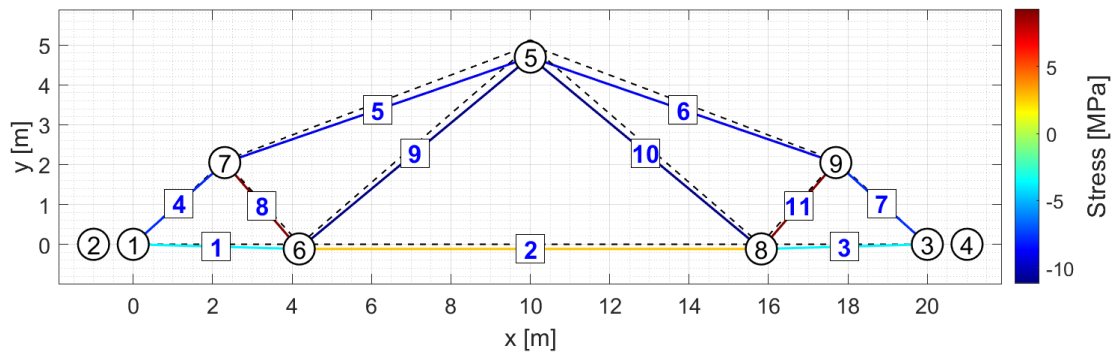


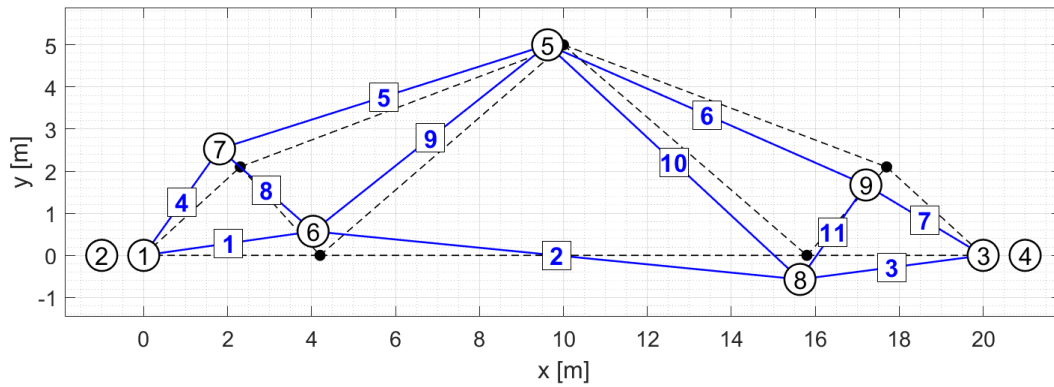
Figure 11: Static deformation of structural steel bridge scaled by a factor of 250. Solid lines show the deformed structure, while dashed lines show the undeformed structure.

3.1.2 Dynamic analysis

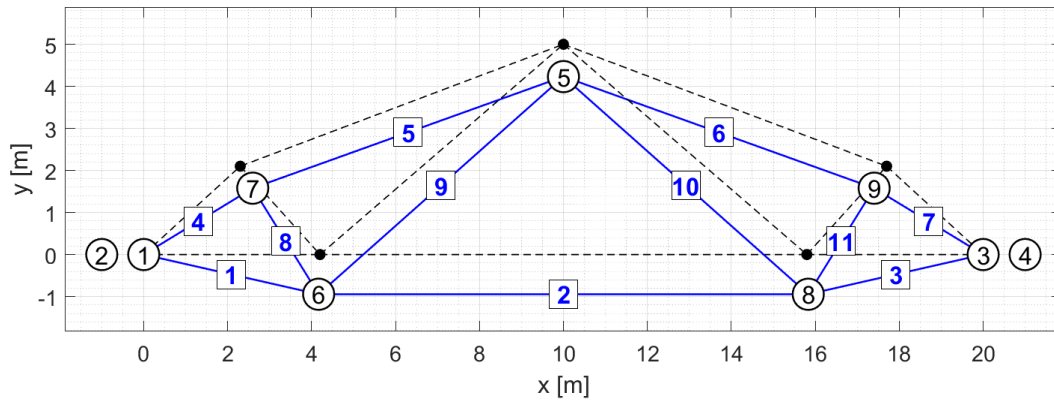
A modal analysis has been performed in order to determine the lowest frequencies of the designed structure. The three lowest natural frequencies obtained during this analysis are presented in table 4, while the corresponding mode shapes are presented by figure 12.

Mode number	1	2	3
Frequency [Hz]	101.87	104.55	134.92

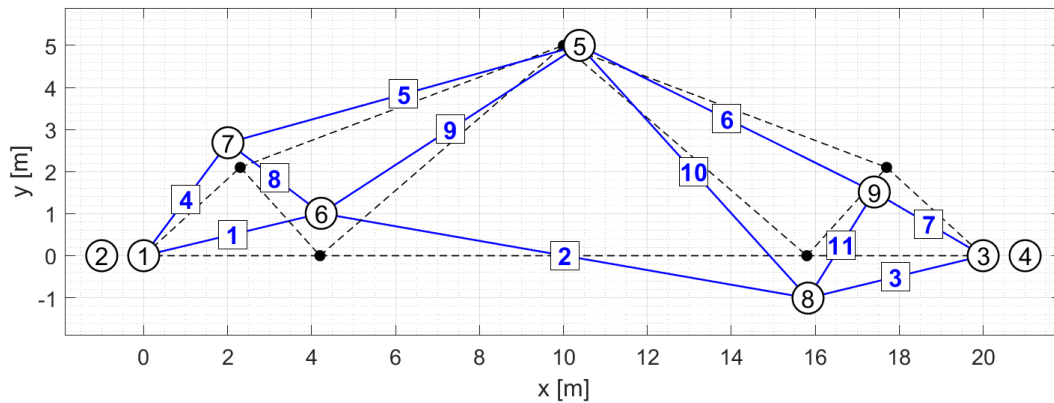
Table 4: First 3 three natural frequencies for the steel bridge.



(a) Mode 1



(b) Mode 2



(c) Mode 3

Figure 12: Mode shapes for the first three modes of the steel structure scaled by an arbitrary factor of 10. Solid lines show the mode shapes, while dashed lines show the undeformed structure.

3.2 Aluminum alloy

The final iteration of the optimizer gave us the geometric parameters presented on figure 13. Furthermore tables 17, 18 and 19 in appendix A.2 present the nodal coordinates, connectivity table and cross-sectional areas for the structure. We found that the yield stress for the aluminum alloy was $\sigma_y = 95$ MPa [2]. The total amount of aluminum used to build this truss is 0.299 m^3 , which fully meets the volume requirement.

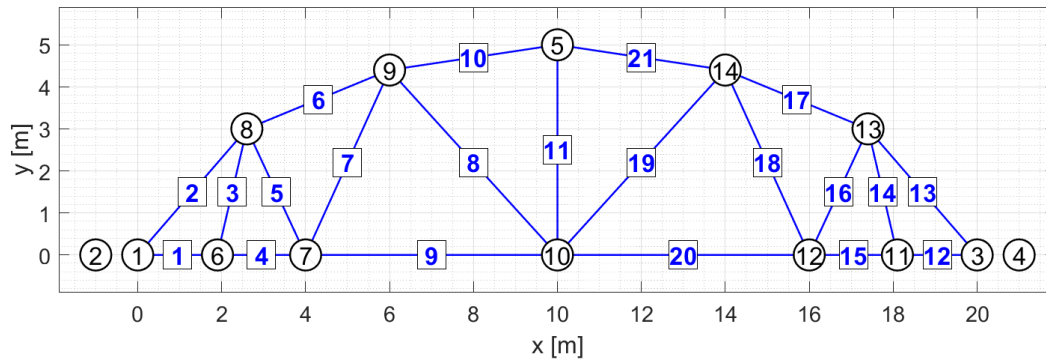


Figure 13: Optimized structure for the aluminum bridge.

3.2.1 Static analysis

Table 5 shows the displacements found for individual nodes, while table 6 shows the stresses found for individual elements. Due to symmetry, results are only shown for the left side of the bridge. Results are not shown for nodes 1 and 2, as they are constrained in both degrees of freedom. A graphical representation of the deformation is shown on figure 14.

Node number	5	6	7	8	9	10
Horizontal displacement [mm]	-0.000	-0.046	-0.099	0.014	0.098	-0.000
Vertical displacement [mm]	-4.029	-0.392	-0.801	-0.406	-1.409	-2.876
Total displacement [mm]	4.029	0.394	0.807	0.406	1.413	2.876

Table 5: Static displacements of nodes 5 to 10 in the aluminum bridge

Element	1	2	3	4	5	6	7	8	9	10	11
Stress [MPa]	-1.68	-5.17	0.00	-1.74	6.47	-5.70	-6.73	11.83	1.15	-8.26	-15.91

Table 6: Static stresses for elements 1 to 11 in the aluminum bridge

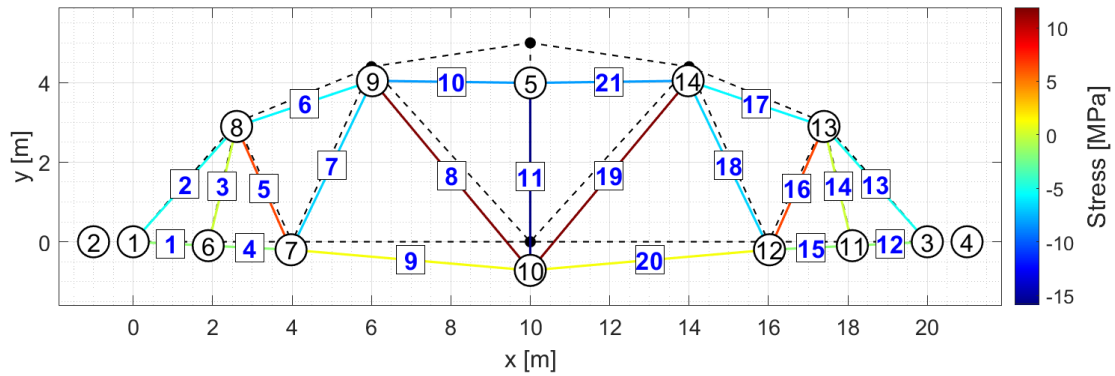


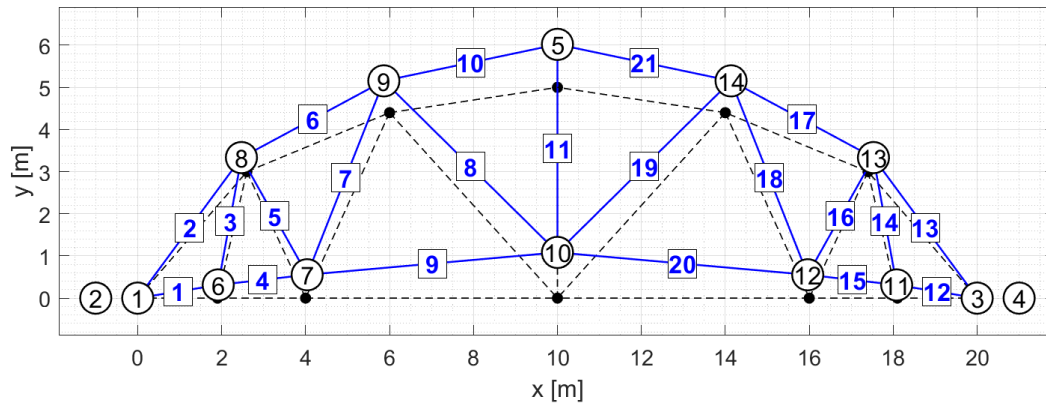
Figure 14: Static deformation of aluminum alloy bridge scaled by a factor of 250. Solid lines show the deformed structure, while dashed lines show the undeformed structure.

3.2.2 Dynamic Analysis

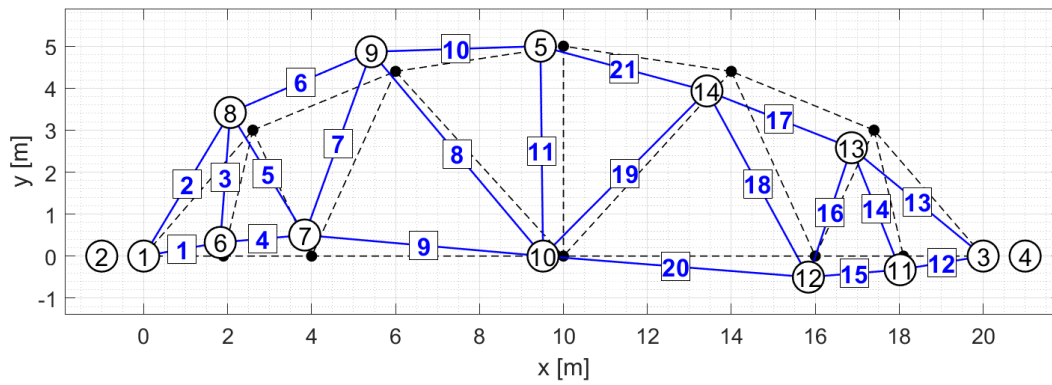
As with the other structures, the dynamic response of the structure have been analyzed using modal analysis. Table 7 presents the three lowest natural frequencies of the structure, while the corresponding mode shapes are presented by figure 15.

Mode number	1	2	3
Frequency [Hz]	73.27	76.09	125.10

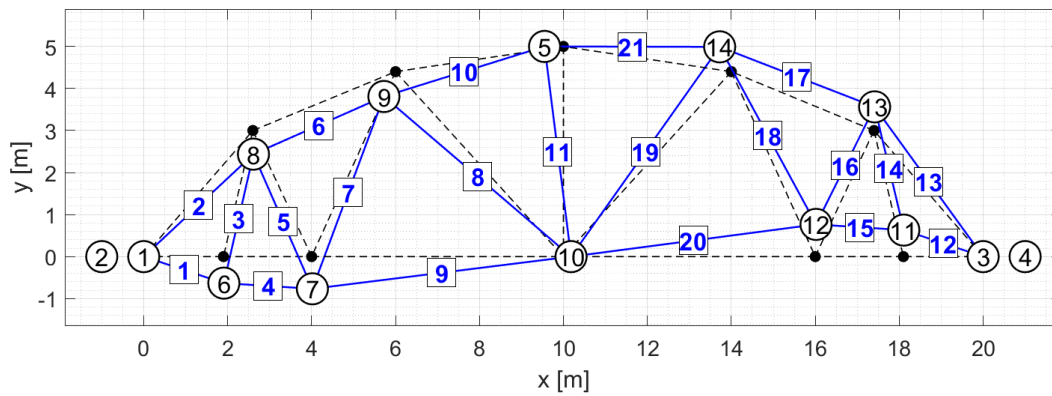
Table 7: First three natural frequencies for the aluminum bridge.



(a) Mode 1



(b) Mode 2



(c) Mode 3

Figure 15: Mode shapes for the first three modes of the aluminum structure scaled by an arbitrary factor of 10. Solid lines show the mode shapes, while dashed lines show the undeformed structure.

3.3 Engineered wood

The final iteration of the optimizer gave us the geometric parameters presented on figure 16. Furthermore tables 20 and 21 in appendix A.2 present the nodal coordinates, connectivity table and cross-sectional areas for the structure. We found that the yield stress for Engineered wood was $\sigma_y = 40$ MPa [2]. The total amount of wood used to build this truss is 0.580 m^3 , which fully meets the volume requirement.

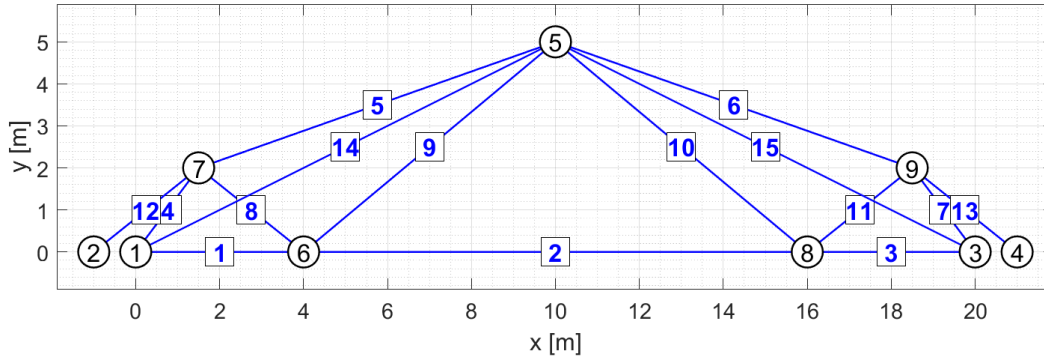


Figure 16: Optimized structure for the wood bridge

3.3.1 Static analysis

Table 8 shows the displacements found for individual nodes, while table 9 shows the stresses found for individual elements. Due to symmetry, results are only shown for the left side of the bridge. The results are not shown for nodes 1 and 2, as they are fully constrained.

Node number	5	6	7
Horizontal displacement [mm]	-0.000	-0.491	0.108
Vertical displacement [mm]	-6.781	-2.670	-0.500
Total displacement [mm]	6.781	2.715	0.512

Table 8: Static displacements of wooden bridge.

Element number	1	2	4	5	8	9	12	14
Stress [MPa]	-1.47	0.98	-1.61	-2.91	3.33	-3.46	-0.85	-3.25

Table 9: Static stresses in the wooden bridge.

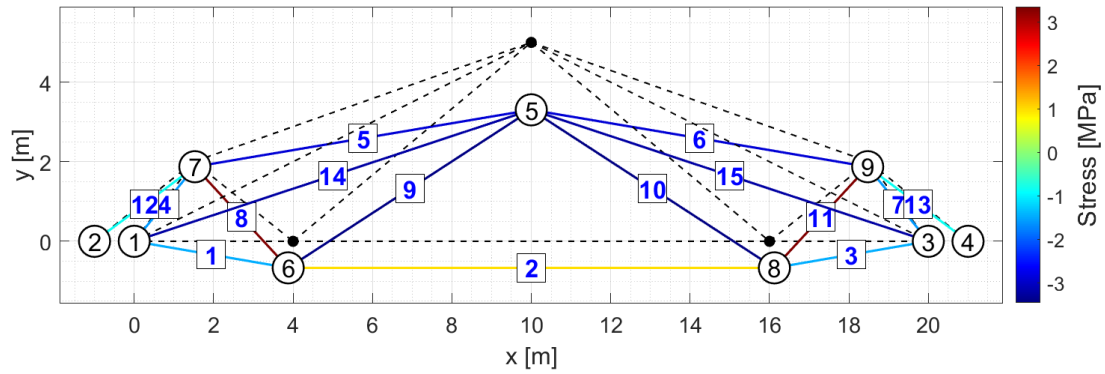


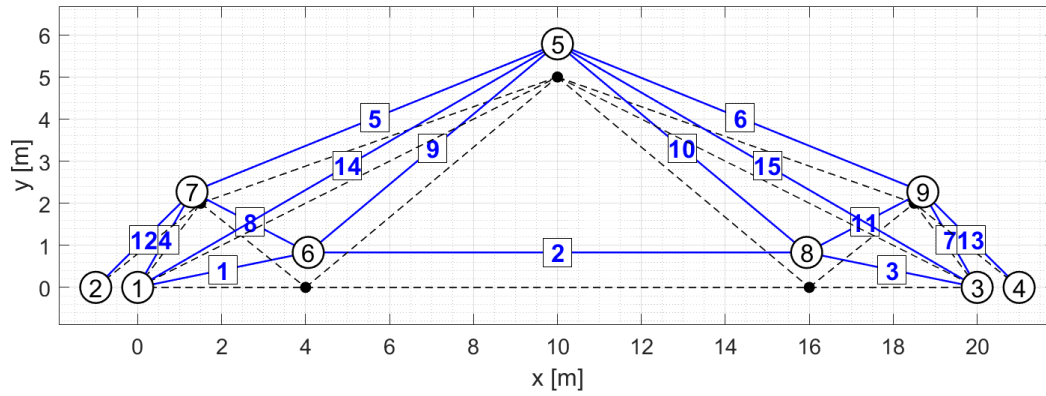
Figure 17: Static deformation of engineering wood bridge scaled by a factor of 250. Solid lines show the deformed structure, while dashed lines show the undeformed structure.

3.3.2 Dynamic Analysis

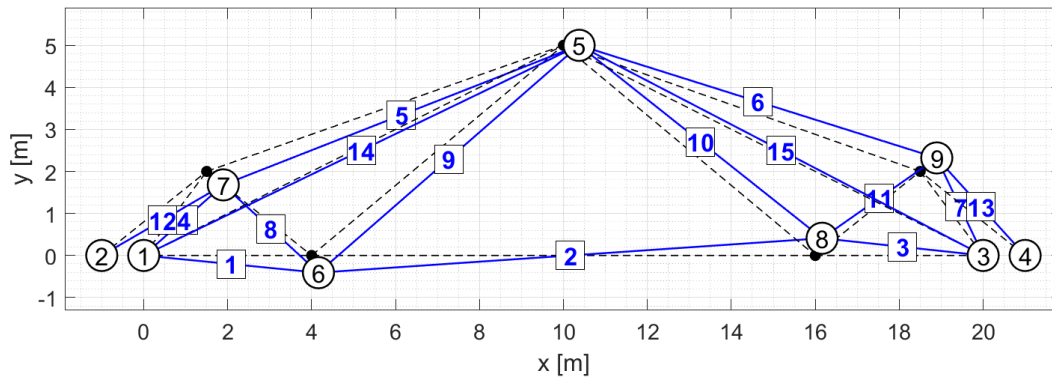
As with the other structures, the dynamic response of the structure have been analyzed using modal analysis. Table 10 presents the three lowest natural frequencies of the structure, while the corresponding mode shapes are presented by figure 18.

Mode number	1	2	3
Frequency [Hz]	98.66	99.11	124.56

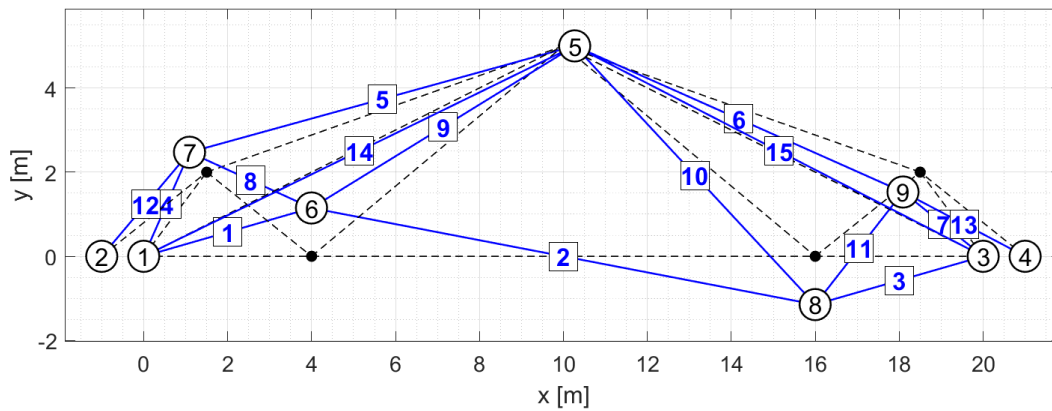
Table 10: First 3 eigenfrequencies for the wooden bridge.



(a) Mode 1



(b) Mode 2



(c) Mode 3

Figure 18: Mode shapes for the first three modes of the wooden structure scaled by an arbitrary factor of 4. Solid lines show the mode shapes, while dashed lines show the undeformed structure.

4 Discussion

4.1 Structural steel

From table 3, it can be deduced that the highest stress (in absolute value) reached in the structure is 11.18 MPa. By comparing this value to the yield stress of structural steel ($\sigma_y = 250$ MPa), it is clear that all the elements remain in the elastic regime. Quantified one obtains a safety factor of 22.4, which is very high suggesting that the designed truss is safe in terms of strength.

Regarding displacements, it can be deduced from table 2 that the maximum nodal displacement is 1.212 mm. This suggests that all the nodes verify the maximum displacement requirement of 40 mm, which means that the structure is stiff enough to be used safely according to the requirements.

For the dynamic analysis, it can be observed that the natural frequencies of the structure are high relatively to the requirements. Indeed, table 4 shows that the lowest resonance frequency is 101.87 Hz which is twice higher the minimum recommended threshold of 50 Hz and ten times higher the minimum required threshold of 10 Hz. Since typical dynamic loads applied on a bridge, such as human walking, vehicle passing or wind gusts, are usually in a low-frequency range, this suggests that the truss will be difficult to excite, preventing it to start oscillating under loadings, thus ensuring stability.

4.2 Aluminum alloy

From table 6, it can be deduced that the highest stress (in absolute value) reached in the structure is 15.91 MPa. By comparing this value to the yield stress of aluminum alloy ($\sigma_y = 95$ MPa), it is clear that all the elements remain in the elastic regime and it obtained that the safety factor is 6.0, which is still very high suggesting that the designed truss is safe in term of strength.

Regarding displacements, it can be deduced from table 5 that the maximum nodal displacement is 4.029 mm. This suggests that all the nodes verify the maximum displacement requirement of 40 mm, which means that the structure is stiff enough to be used safely according to the requirements.

For the dynamic analysis, it can be observed that the natural frequencies of the structure are high relatively to the requirements. Indeed, table 7 shows that the lowest resonance frequency is 73.27 Hz which is 46.5 % higher the minimum recommended threshold of 50 Hz and more than 7 times higher the minimum required threshold of 10 Hz. Similarly to the steel truss, this suggests that the truss will be difficult to excite preventing it to start oscillating under loadings, thus ensuring stability.

4.3 Engineered wood

From table 9, it can be deduced that the highest stress (in absolute value) reached in the structure is 3.46 MPa. By comparing this value to the yield stress of engineered wood ($\sigma_y = 40$ MPa), it is clear that all the elements remain in the elastic regime and it obtained that the

safety factor is 11.6, which is very high suggesting that the designed truss is safe in term of strength.

Regarding displacements, it can be deduced from table 8 that the maximum nodal displacement is 6.781 mm. This suggests that all the nodes verify the maximum displacement requirement of 40 mm, which means that the structure is stiff enough to be used safely according to the requirements.

For the dynamic analysis, it can be observed that the natural frequencies of the structure are high relatively to the requirements. Indeed, table 10 shows that the lowest resonance frequency is 98.66 Hz which is almost twice higher the minimum recommended threshold of 50 Hz and almost ten times higher the minimum required threshold of 10 Hz. Similarly to the two previous trusses, this suggests that the truss will be difficult to excite preventing it to start oscillating under loadings, thus ensuring stability.

4.4 Comparison

A summary of the results are presented in table 11.

Truss	Structural steel	Aluminum alloy	Engineered wood
Max stress [MPa]	11.18	15.91	3.46
Safety factor [-]	22.3	6.0	11.6
Max displacement [mm]	1.212	4.029	6.781
Volume of materials [m ³]	0.196	0.299	0.580
First frequency [Hz]	101.87	73.27	98.66

Table 11: Summary of the main results.

By comparing the results of the different trusses, it can be concluded that the one made of steel is the one fulfilling the different requirements with the most margin, by having the highest safety factor and first natural frequency, and having the lowest maximum displacement and amount of materials used. Then it is less clear which one of the two other trusses is performing the best, since aluminum has a better displacement and amount of material used, while wood has a better safety factor and first natural frequency.

Beyond the good results that were obtained, it is important to keep in mind that the model used is based on several assumptions, as stated in section 2.1.1. Indeed, the results could be expected to be slightly worse in reality, since friction in pins could induce some additional shear stress and this, along with bars self weight, would increase the overall stress within bars, thus also increasing nodal displacement. Considering self weight would also enhance the advantages and disadvantages of the different materials considered.

Additionally, in a real case scenario, the bridges would be subject to dynamic loading, resulting from people or vehicle crossings and natural factors such as wind. This would make the analysis somewhat more complex and would make the structure subject to fatigue.

All in all, these simplifications do not mean that the structure is actually weak, but highlight the fact that some safety margins are required to really ensure strength and stability. Fortunately, all the three designs have significant safety margins on all parameters. On the other hand, the

safety margins highlight the fact that there is room for change in the trusses in order to favour some of the requirements to the detriment of others. E.g. one could reduce the volume of used material further, and thereby the price, at the cost of a decrease in structural stiffness.

Regarding the mode shapes, it can be observed that similar behaviors are obtained between the different trusses, despite that their configuration varies. However, it is important to notice that surprisingly the first two mode shapes of the steel structure seems to be inverted compared to the aluminum or wooden bridges, although the steel and wooden bridges are quite similar. Apart from that, the mode shapes interestingly seem to fit some sinusoidal shapes, similarly to the case of a single beam simply supported at its ends. Indeed, one of the mode shapes looks like a half-period sinusoid, while the two others, very similar to one another, look like a full-period sinusoid.

Interestingly, all optimized designs tend to converge toward a similar overall shape—arch-like structures—which have proven to be the most effective for reducing structural compliance and maximizing the first modal frequency. During the optimization process, various alternative designs were explored. In particular, some attempts focused on better utilizing the outermost ground points. However, the approach of developing tower-like structures on the sides to suspend the central part of the bridge proved ineffective. This is mainly because the ground points are too close to each other, resulting in low-stiffness configurations with high displacements.

5 Conclusion

In conclusion, it was seen that the designed trusses made of different materials meet all the four requirements, described in section 1.1, with a relatively high safety margin. It was also found that the truss made of structural steel tends to overall perform the best when comparing to the aluminum and wooden bridges. Additionally, it was discussed that the different assumptions that were made in this analysis could lead to slightly better results than in a real situation. Nevertheless, the relatively high safety margins obtained play an important role in ensuring strength and stability despite these unaccounted complexities.

6 References

- [1] Stefano Burzio. *Dynamic Finite Element Analysis of Structures*. Lecture slides, EPFL, ME-473. Spring Semester 2025.
- [2] The Engineering ToolBox. *Young's Modulus of Elasticity – Values for Common Materials*. https://www.engineeringtoolbox.com/young-modulus-d_417.html. [online] Accessed: 9 Apr 2025. 2003.
- [3] Jon Juel Thomsen. *Vibrations and stability*. Vol. 3. Springer, 2021.

A Details on FE-structures.

A.1 Optimization reference geometries

In this section, the final 3 models used for the different material are reported.

A.1.1 Reference for steel bridge: Warren bridge.

The first and easiest design was inspired by the Warren bridge design.

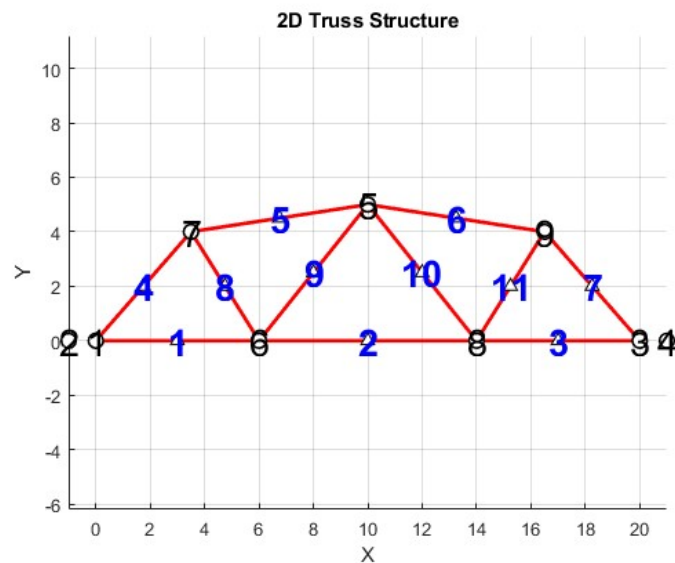


Figure 19: Warren reference bridge

It's important to note that the initial elements' areas were defined as a random value that lies inside the areas boundary of all the materials (user-defined variable).

Node	1	2	3	4	5	6	7	8	9
x [m]	0	-1	20	21	10	6	3.5	14	16.5
y [m]	0	0	0	0	5	0	4	0	4

Table 12: Node positions for the reference Warren bridge

A.1.2 Reference for aluminum bridge: Pratt bridge.

The second design is based on the Pratt bridge design.

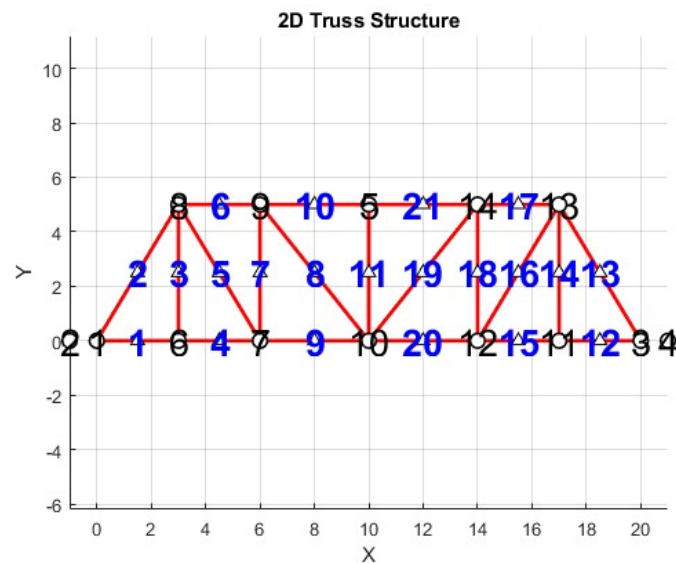


Figure 20: Pratt reference bridge

Node	1	2	3	4	5	6	7	8	9	10	11	12	13	14
x [m]	0	0	-1	0	20	3	6	3	6	10	17	14	17	14
y [m]	2.6	3	6	4.4	10	0	0	5	5	0	0	0	5	5

Table 13: Node positions for the reference Pratt bridge

A.1.3 Reference for wooden bridge: Increased stiffness Warren bridge.

The Warren bridge inspired the last design, but it was reinforced to increase stiffness and reduce displacement, as it was meant for low Young's modulus materials, such as the last proposed material: wood. In particular, beams 14 and 15 are meant to connect the ground nodes directly with the one with the applied load. These beams are meant to lie on parallel planes with the rest of the beams, allowing for no connection between different beams. This was considered to be compliant with the requirements of the problem.

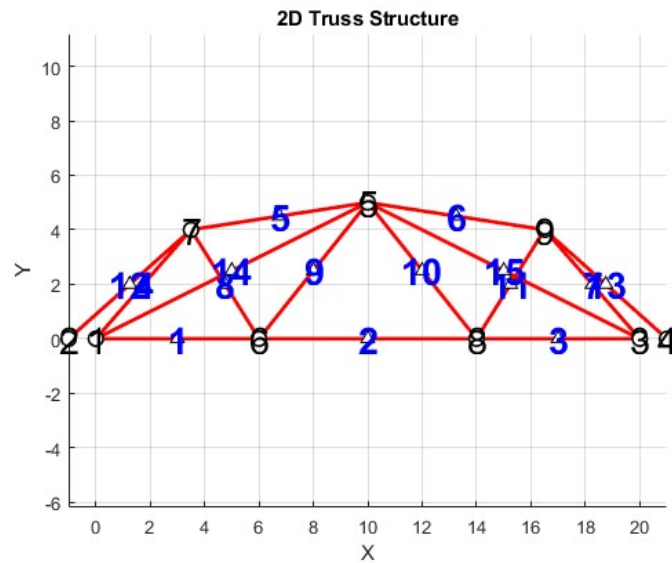


Figure 21: High stiffness warren reference bridge

Node	1	2	3	4	5	6	7	8	9
x [m]	0	-1	20	21	10	6	3.5	14	16.5
y [m]	0	0	0	0	5	0	4	0	4

Table 14: Node positions for the reference increased stiffness Warren bridge

A.2 Optimized bridge structures

Tables 15 and 16 respectively presents the nodal coordinates and the connectivity tables for the structural steel bridge structure. Table 16 also presents the cross-sectional areas of the individual elements.

Node	1	2	3	4	5	6	7	8	9
x [m]	0	-1	20	21	10	4.2	2.3	15.8	17.7
y [m]	0	0	0	0	5	0	2.1	0	2.1

Table 15: Nodal coordinates for the steel bridge.

Element	1	2	3	4	5	6	7	8	9	10	11
Node 1	1	6	8	1	7	5	9	7	6	5	8
Node 2	6	8	3	7	5	9	3	6	5	8	9
Area [cm ²]	46	20	46	47	44	44	47	16	15	15	16

Table 16: Connectivity and cross section table for the steel bridge

Tables 17, 18, and 19 respectively presents the nodal coordinates and the connectivity tables for the structural steel bridge structure. Tables 18 and 19 also present the cross-sectional areas of the individual elements.

Node	1	2	3	4	5	6	7	8	9	10	11	12	12	14
x [m]	0	0	-1	0	20	0	21	0	10	5	1.9	0	4	0
y [m]	2.6	3	6	4.4	10	0	18.1	0	16	0	17.4	3	14	4.4

Table 17: Node positions for the aluminum bridge.

Element	1	2	3	4	5	6	7	8	9	10	11	12
Node 1	1	6	1	8	8	6	6	7	8	7	8	9
Node 2	11	3	13	13	11	11	12	13	12	13	14	14
Area [cm²]	57	64	23	55	23	53	22	20	25	61	22	57

Table 18: Connectivity and cross section table for elements 1 to 12 for the aluminum bridge

Element	13	14	15	16	17	18	19	20	21	22	23
Node 1	9	7	9	10	7	10	9	5	5	2	3
Node 2	12	14	10	12	10	14	5	2	1	3	4
Area [cm²]	64	23	55	23	53	22	20	25	61	20	20

Table 19: Connectivity and cross section table for elements 13 to 23 for the aluminum bridge

Tables 20 and 21 respectively presents the nodal coordinates and the connectivity tables for the structural steel bridge structure. Table 21 also present the cross-sectional areas of the individual elements.

Node	1	2	3	4	5	6	7	8	9
x [m]	0	-1	20	21	10	4	1.5	16	18.5
y [m]	0	0	0	0	5	0	2	0	2

Table 20: Node positions for the wood bridge

Element	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Node 1	1	6	8	1	7	5	9	7	6	5	8	2	4	1	5
Node 2	6	8	3	7	5	9	3	6	5	8	9	7	9	5	3
Area [cm²]	88	34	88	72	78	78	72	32	30	30	32	93	93	74	74

Table 21: Connectivity and cross section table for the wooden bridge